Online Appendix to "Separating the Wheat from the Chaff"

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Section OA.1 characterizes the "min" selection rule. Section OA.2 provides a direct proof of Lemma 2 of the paper, verifying the expert's incentive to report truthfully. Section OA.3 studies how the accuracy of the judge's signal affects the judge's equilibrium value, using the dispersive order to compare noise distributions. Section OA.4 studies equilibrium pandering with an uninformed quack and smooth noise. Section OA.5 studies sequential communication in the "polite talk" game. Section OA.6 characterizes the judge's preferred equilibrium in the model with one speaker and an outside option. Section OA.7 presents numerical evidence that the information generated by a partition equilibrium need not be Blackwell-dominated by that generated by a truth-telling equilibrium. Sections and results of this appendix are cited throughout the paper. The notation of the paper is used throughout.

OA.1 Min Rule

Define the class of "min" selection rules as follows: a selection rule ϕ belongs to Φ^{min} if, when both messages are consistent — so that $|m_1 - s|, |m_2 - s| \leqslant \bar{\varepsilon}$ — and message i is the least extreme — $|m_i| < |m_{-i}|$,

$$\begin{aligned}\phi(m_1, m_2, s) = \phi_{min}(\min\{|m_1|, |m_2|\})\delta_i \\ + (1 - \phi_{min}(\min\{|m_1|, |m_2|\}))\delta_{-i},\end{aligned}$$

for some continuous $\phi_{\min} : [0,1] \to [0,1]$. When $|m_1| = |m_2|$, each speaker is selected with probability $1/2$.

Lemma OA.1.1. *There exists a unique continuous function $\phi_{\min} : [0,1] \to [0,1]$ that can be completed to a selection rule in $\Phi^{ind} \cap \Phi^{min}$. The function ϕ_{min} is strictly decreasing over $[0,1]$ and satisfies $0 < \phi_{min}(m) < 1/2$ for every m in $[0,1]$.*

Proof. Put $e = \bar{\epsilon}$, $C = e - e^2/3$, and $f = \phi_{\min}$. Define

$$W(m,x) = \left(1 - \frac{|m-x|}{2e}\right)_+ + \left(1 - \frac{m+x}{2e}\right)_+,$$

$$A(m) = \int_0^m W(m,x)\,\mathrm{d}x, \quad B(m) = \int_m^1 W(m,x)\,\mathrm{d}x.$$

Indifference requires

$$B(m)f(m) = C - A(m) + \int_0^m W(m,x)f(x)\,\mathrm{d}x. \tag{1}$$

Since $B > 0$ on $[0,1)$, this Volterra equation has a unique continuous solution on every compact subinterval of $[0,1)$. These solutions agree on their common domains. In particular,

$$f(0) = \begin{cases} 1/2 - e/6, & e \le 1/2, \\ (e - e^2/3)/(2 - 1/(2e)), & e > 1/2, \end{cases} \qquad 0 < f(0) < \frac{1}{2}.$$

The solution is continuously differentiable and is twice differentiable away from the finitely many changes in the kernel's formula.

We first establish continuity at one. Suppose $e < 1$ and extend f evenly to $(-1,1)$. For m sufficiently close to one, $A(m) = e$ and $B(m) = (1-m) - (1-m)^2/(4e)$. Differentiating (1) twice yields

$$B(m)f''(m) + (2B'(m) - 1)f'(m) = \frac{f(|m-2e|)}{2e}.$$

The argument $|m - 2e|$ stays away from one, so the right side is bounded. An integrating factor gives, for a constant c,

$$f'(m) = \frac{c - 2\int_m^1 (1-t)^2 f(|t-2e|)\,\mathrm{d}t}{(1-m)^3(4e-1+m)}. \tag{2}$$

Thus $f(m) = c/[8e(1-m)^2] + o((1-m)^{-2})$. Integrating the indifference equation from zero to m and using symmetry of W gives

$$\int_0^m f(x)\int_m^1 W(x,y)\,\mathrm{d}y\,\mathrm{d}x = mC - \frac{1}{2}\int_0^m\int_0^m W(x,y)\,\mathrm{d}y\,\mathrm{d}x. \tag{3}$$

The right side tends to zero. The left side tends to $c/(8e)$: on any fixed interval away from one its contribution vanishes, and near $(1,1)$ the kernel converges uniformly to $W(1,1) = 1$. The displayed asymptotic for f then gives this limit. Hence $c = 0$. Equation (2) now implies that f' has a finite limit, and f extends continuously to one. Differentiating (1) once and taking limits gives

$$f(1) = \frac{1}{4e}\int_{1-2e}^1 f(x)\,\mathrm{d}x, \qquad f'(1) = -\frac{f(|1-2e|)}{6e}. \tag{4}$$

If $e = 1$, direct solution of (1) instead gives

$$f(m) = \frac{2}{3}\left(1 - \frac{3}{(3-m)^2}\right).$$

This function is positive and strictly decreasing, with $f(1) = 1/6$. We henceforth take $e < 1$.

Write $D = A + B$. Direct integration gives

$$D(m) = 2e - \frac{(m+2e-1)_+^2 + (-m+2e-1)_+^2}{4e}, \qquad D' \le 0, \quad D'' \le 0.$$

Also $A' = 2 - 2m/e$ for $m < e$, $A' = 0$ for $m \ge e$, and $B' \le 0$. For $m < e$, put $F(m) = \int_0^m f(x)\,\mathrm{d}x$. Differentiating (1) gives

$$Bf' = -A' + (2 - m/e - B')f - F/e. \tag{5}$$

For $m \geq e$, differentiation gives

$$B(m)f'(m) = (1 - B'(m))f(m) - \frac{1}{2e}\int_{m-2e}^{m} f(x)\,\mathrm{d}x. \tag{6}$$

Suppose the global minimum of f is nonpositive, and take the first point m attaining it. This point is positive. If $m < e$, then $F(m) > mf(m)$, so (5) implies

$$B(m)f'(m) < -A'(m) + (A'(m) - B'(m))f(m) < 0.$$

If $e \leq m < 1$, all arguments inside the integral in (6) have absolute value below m, apart from endpoints. Thus that equation gives $B(m)f'(m) < -B'(m)f(m) \leq 0$. Both conclusions contradict an interior minimum. If $m = 1$, the first identity in (4) says that $f(1)$ is one-half of an average of values larger than $f(1)$, which also contradicts $f(1) \leq 0$. Hence $f > 0$.

If the global maximum is at least $1/2$, take its first point m. Again $m > 0$. For $m < e$, (5) gives

$$B(m)f'(m) > A'(m)(2f(m) - 1) - D'(m)f(m) \geq 0.$$

For $e \leq m < 1$, (6) gives $B(m)f'(m) > -B'(m)f(m) \geq 0$. An interior maximum is impossible. An endpoint maximum is excluded by (4), since it would be at most one-half of itself. Therefore $0 < f < 1/2$ on $[0, 1]$.

It remains to prove monotonicity. Put $q = f'$ and

$$L(m, x) = \left(1 - \frac{|m - x|}{2e}\right)_+ - \left(1 - \frac{m + x}{2e}\right)_+ \geq 0 \qquad (0 \leq x \leq m).$$

Integration by parts in the differentiated indifference equation gives

$$B(m)q(m) = A'(m)(2f(m) - 1) - D'(m)f(m) + \int_0^m L(m, x)q(x)\,\mathrm{d}x.$$

At zero, $q(0) = (4f(0) - 2)/B(0) < 0$. On $[0, (1 - 2e)_+]$ we have $D' = 0$. Since $f < 1/2$, a first zero of q on this interval would make the right side negative. Thus $q < 0$ there.

On the remaining interval, $D'' \leq 0$ and, whenever $m \geq e$, $B'' = -1/(2e)$. Differentiating (5) and (6), respectively, gives

$$\begin{cases} Bq' + (2B' - (2 - m/e))q = 2/e - (4/e + D'')f > 0, & m < e, \\ Bq' + (2B' - 1)q = f(|m - 2e|)/(2e) > 0, & m \geq e. \end{cases}$$

These identities hold away from the changes in formula and almost everywhere throughout the interval. The coefficient of q agrees at $m = e$. Since (4) gives $q(1) < 0$, integration backward with a positive integrating factor implies $q(m) < 0$ throughout this remaining interval as well. This proves strict decrease. The continuous solution satisfies (1) also at one by taking limits. Uniqueness on $[0, 1)$ and continuity give uniqueness on $[0, 1]$. □

OA.2 Proof of Lemma 2

Fix a speaker identity and write $p(m, m', s)$ for the probability that this speaker is selected after reporting m, when the other report is m'. For each report m, define its *consistency interval* by $I(m) = [m - \bar{\epsilon}, m + \bar{\epsilon}]$. Write

$$c(s) = \frac{1}{2}\int_{-1}^{1} \mathbf{1}\{|m' - s| \leq \bar{\epsilon}\}\,\mathrm{d}m'$$

for the probability that a uniformly drawn opposing report is consistent with s.

Quack indifference implies that, for every m in $[-1,1]$,

$$\int_{I(m)}\int_{-1}^{1}\frac{p(m,m',s)\mathbf{1}\{|m'-s|\leq\bar{\epsilon}\}}{4\bar{\epsilon}}\,\mathrm{d}m'\,\mathrm{d}s=u,$$

where u is this identity's quack payoff. No restriction on the dependence of p on s is imposed.

When the expert observes ω, truthful reporting gives

$$V_E(\omega,\omega)=\int_{I(\omega)}\frac{1-c(s)}{2\bar{\epsilon}}\,\mathrm{d}s+u.$$

For any alternative report m, his payoff is at most

$$\int_{I(\omega)}\frac{1-c(s)}{2\bar{\epsilon}}\,\mathrm{d}s+\int_{I(\omega)\cap I(m)}\int_{-1}^{1}\frac{p(m,m',s)\mathbf{1}\{|m'-s|\leq\bar{\epsilon}\}}{4\bar{\epsilon}}\,\mathrm{d}m'\,\mathrm{d}s.$$

The first term allows the expert to win whenever the quack is inconsistent, including observations at which both reports are inconsistent. If the quack is consistent and the expert is not, the expert loses. Subtracting gives

$$V_E(\omega,\omega)-V_E(m,\omega)\geq\int_{I(m)\setminus I(\omega)}\int_{-1}^{1}\frac{p(m,m',s)\mathbf{1}\{|m'-s|\leq\bar{\epsilon}\}}{4\bar{\epsilon}}\,\mathrm{d}m'\,\mathrm{d}s\geq 0.$$

Thus truth-telling is optimal for every state, whatever the selection rule after two inconsistent reports.

Under the max and min rules, each speaker has positive selection probability whenever both reports are consistent. For $m\neq\omega$, $I(m)\setminus I(\omega)$ has positive length, and $c(s)>0$ almost everywhere on this difference. The first inequality therefore has a positive right-hand side. This proves uniqueness for these rules. □

OA.3 Signal Accuracy

In the benchmark, the judge selects the expert with probability $1-\bar{\epsilon}/2+\bar{\epsilon}^2/6$, which increases with signal accuracy. This payoff differs from the probability of learning the state with certainty in Proposition 2 of the paper.

Proposition OA.3.1 proposes a more general version of this result. Maintaining the extreme definitions of expert and quack — fully informed versus fully uninformed, respectively — and generalizing the class of judge's signal to that considered in our general model, the proposition formalizes a sense in which the judge's value is increasing in the accuracy of her private signal. To that end, we use Lehmann's (1988, Theorem 5.2) ordering of experiments by effectiveness. For location families, its quantile characterization is the *dispersive order*: writing H_i for the cdf of ϵ_i and $H_i^{-1}(p):=\inf\{x:H_i(x)\geqslant p\}$, the noise ϵ_1 is less dispersed than ϵ_2 (denoted $\epsilon_1\leqslant_{\text{disp}}\epsilon_2$) if[1]

$$H_1^{-1}(p)-H_1^{-1}(q)\leqslant H_2^{-1}(p)-H_2^{-1}(q),\qquad\text{for all } 0<q<p<1.$$

Fix the state distribution G and the identity priors p_1 and $p_2=1-p_1$. For any noise density h, $v(h)$ denotes the quack's value in the corresponding truth-telling game.

[1]Within any scale family — the uniform family of the benchmark, say, or the family $h_\sigma(\cdot)=h(\cdot/\sigma)/\sigma$ generated by any fixed density — a smaller scale parameter means a dispersively smaller noise.

Proposition OA.3.1. *Let ϵ_1 and ϵ_2 be noise variables with densities h_1 and h_2, such that $\epsilon_1 \leqslant_{\text{disp}} \epsilon_2$ and h_2 is log-concave. Then $v(h_1) \leqslant v(h_2)$. Thus greater dispersion weakly increases the quack's value $v(h)$ and weakly decreases the judge's value $1 - v(h)$ in the truth-telling game.*

Proof. The proof exploits the two-point structure of the judge's inference problem: conditional on the pair of reports and a truthful expert, the state is one of the two reports, so the judge faces a family of binary tests between two possible locations. Throughout, H_i denotes the cdf of ϵ_i and $H_i^{-1}(p) = \inf\{x : H_i(x) \geqslant p\}$ its left-continuous inverse; since densities exist, each H_i is continuous, so that $H_i(H_i^{-1}(p)) = p$ for p in $(0, 1)$.

We first claim that, for every shift t in $\mathbf{R}$ and all weights $c_0, c_1 \geqslant 0$,

$$\int_{\mathbf{R}} \min\{c_0 h_1(s), c_1 h_1(s-t)\}\,\mathrm{d}s \leqslant \int_{\mathbf{R}} \min\{c_0 h_2(s), c_1 h_2(s-t)\}\,\mathrm{d}s. \tag{7}$$

If $t = 0$ or $c_0 c_1 = 0$, both sides coincide (they equal $\min\{c_0, c_1\}$ and 0, respectively). By the relabeling $(c_0, c_1, t) \mapsto (c_1, c_0, -t)$ and a change of variables, assume $t > 0$ and $c_0, c_1 > 0$. Interpret each side as the *Bayes error*, the minimal weighted error for testing $P_0 : s \sim h_i$ against $P_t : s \sim h_i(\cdot - t)$: for any measurable acceptance region A ("decide P_t"),

$$c_0 P_0(A) + c_1 P_t(A^{\complement}) \geqslant \int \min\{c_0 h_i(s), c_1 h_i(s-t)\}\,\mathrm{d}s =: e_i(c_0, c_1, t),$$

with equality for the Bayes region $A_i^* = \{s : c_1 h_i(s-t) > c_0 h_i(s)\}$.

First, since h_2 is log-concave and $t > 0$, the likelihood ratio $s \mapsto h_2(s-t)/h_2(s)$ is nondecreasing on the set where it is defined, equal to 0 below the support of $h_2(\cdot - t)$, and equal to $+\infty$ above the support of h_2. Hence, A_2^* is, up to sets that are null under both hypotheses, a right half-line (k_2, ∞). Put $p = H_2(k_2)$ and first suppose $0 < p < 1$. The interval support of h_2 gives $k_2 = H_2^{-1}(p)$, so

$$e_2 = c_0(1-p) + c_1 H_2\big(H_2^{-1}(p) - t\big).$$

Second, match the false-positive rate in experiment 1 with a cut-off test: let $k_1 := H_1^{-1}(p)$, so that $H_1(k_1) = p$ by continuity, and let $q := H_1(k_1 - t)$. The cut-off test $A := (k_1, \infty)$ in experiment 1 has weighted error $c_0(1-p) + c_1 q$, so that $e_1 \leqslant c_0(1-p) + c_1 q$. Third, the dispersive order converts the type-II comparison. If $q = 0$, the comparison below is immediate. Otherwise $0 < q < p < 1$. By definition of the left-continuous inverse, $H_1^{-1}(q) \leqslant k_1 - t$, while $H_1^{-1}(p) = k_1$. Hence,

$$H_2^{-1}(p) - H_2^{-1}(q) \geqslant H_1^{-1}(p) - H_1^{-1}(q) \geqslant t,$$

the first inequality by the dispersive order. Rearranging and applying the nondecreasing map H_2, then continuity of H_2,

$$q = H_2\big(H_2^{-1}(q)\big) \leqslant H_2\big(H_2^{-1}(p) - t\big).$$

Combining, $e_1 \leqslant c_0(1-p) + c_1 q \leqslant c_0(1-p) + c_1 H_2(H_2^{-1}(p) - t) = e_2$, which is (7). If p is an endpoint, approximate the Bayes cutoff by interior quantiles. The corresponding cutoff errors for experiment 2 converge to its Bayes error, so the same inequality follows by taking limits.

Fix the two speakers' quack strategies (F_Q^1, F_Q^2) and truthful experts. The two weighted observation laws on $\mathbf{R}^2 \times \mathbf{R}$ are

$$p_2 F_Q^1(\mathrm{d}a) G(\mathrm{d}b) h(s-b)\,\mathrm{d}s \qquad \text{and} \qquad p_1 G(\mathrm{d}a) F_Q^2(\mathrm{d}b) h(s-a)\,\mathrm{d}s.$$

Let $\mu := F_Q^1 \otimes G + G \otimes F_Q^2$ on $\mathbf{R}^2$, and let $\pi_1 := \mathrm{d}(F_Q^1 \otimes G)/\,\mathrm{d}\mu$ and $\pi_2 := \mathrm{d}(G \otimes F_Q^2)/\,\mathrm{d}\mu$ be Borel versions of the densities. Disintegrating, the judge's Bayes error under noise density h_i is

$$B_{h_i}(F_Q^1, F_Q^2) = \int_{\mathbf{R}^2} e_i\Big(p_2\pi_1(a,b), p_1\pi_2(a,b), a-b\Big)\,\mu(\mathrm{d}a, \mathrm{d}b),$$

where $e_i(c_0, c_1, t)$ is the two-point testing error derived above (after translating s by b; the shift between the two hypothesized signal locations is $t = a - b$). By the preceding comparison, the integrand is weakly smaller under h_1 than under h_2, for μ-almost every (a, b) — including pairs with $a = b$, or with a vanishing weight, where the two coincide. Since μ and (π_1, π_2) do not depend on the noise, integrating proves $B_{h_1}(F_Q^1, F_Q^2) \leqslant B_{h_2}(F_Q^1, F_Q^2)$.

By Theorem 2 (in the main body of the paper), for every Borel noise density the truth-telling game has a value, and

$$v(h) = \max_{(F_Q^1, F_Q^2)} B_h(F_Q^1, F_Q^2).$$

The saddle pair attains $B_h(F_1^*, F_2^*) = v(h)$ (the equilibrium selection rule is Bayes-optimal at the equilibrium pair), and no pair can achieve more, since the completed equilibrium rule holds every pair's payoff below $v(h)$, while B_h is an infimum over rules. Taking maxima yields $v(h_1) \leqslant v(h_2)$. Finally, fixing the experts at truth-telling makes the quack–judge interaction zero-sum, so in every equilibrium of the communication game with truthful experts the judge's payoff equals $1 - v(h)$, and hence $1 - v(h_1) \geqslant 1 - v(h_2)$. □

OA.4 Pandering

This section considers an environment where the expert is perfectly informed about the state ω, the quack is fully uninformed, and the judge observes a signal $s = \omega + \epsilon$. The state has distribution G over $[-1, 1]$, while the noise ϵ has density h. Theorem 2 in the paper establishes the existence of a saddle-point equilibrium with a truthful expert in this environment.

The purpose of this section is to characterize the quack's equilibrium report distribution and the judge's associated selection rule. We first establish that the quack's report law is absolutely continuous with respect to the expert's report law (Section OA.4.1). We then show that, under a full-range likelihood-ratio condition, the equilibrium quack strategy f_Q is unique, common to the two possible quack identities, and symmetric around zero. Under the noise and prior assumptions in Section OA.4.2, this strategy has a positive Lipschitz continuous density. Its density relative to the prior decreases toward the extremes, and the judge's selection cutoff favors the more extreme report.

Proofs are presented in Section OA.4.3.

OA.4.1 Absolute Continuity

We begin with a general implication of Bayes optimality and the quack's conditional equalization condition in Section C.1.2 of the proof of Theorem 2: At any saddle point of the truth-telling game, we have

$$\mathbf{E}[\mathbf{1}\{m_i \in A\} u_i(t_i)] \leq \Gamma(A) \qquad \text{for every Borel } A, \tag{8}$$

where m_i and t_i denote quack i's report and private signal, respectively. Further, $u_i(t_i)$ denotes his equilibrium probability of being selected conditional on that signal.

This condition states that the quack's equilibrium payoff attributable to reports in A cannot exceed the probability with which a truthful expert reports a value in A. Indeed, under the assignment in which i is the quack, integrate the selection probability over observations with m_i in A; on the region where i is selected, its assignment likelihood is no larger than the opposite assignment likelihood. The latter assigns the report of speaker i the expert law Γ, giving the right side. Equalization converts the left side into the expression displayed in (8).

We now specialize this observation to the case studied in this section, in which the expert is fully informed about the state ω in $[-1,1]$, whose distribution is G, while the quack is uninformed. Conditional on the state ω, the judge's signal has density $h(s-\omega)$. Define the average overlap between the judge's signal distributions under two independent draws from G by

$$\beta := \iint \int \min\{h(s-\omega), h(s-\omega')\}\,\mathrm{d}s\,G(\mathrm{d}\omega)G(\mathrm{d}\omega').$$

Proposition OA.4.1. *The average overlap β satisfies $0 < \beta/2 \leq v \leq 1/2$. At any saddle point of the truth-telling game, if quack identity i has payoff $u_i > 0$, then $F_Q^i \ll G$ and*

$$\frac{\mathrm{d}F_Q^i}{\mathrm{d}G} \leq \frac{1}{u_i} \qquad G\text{-almost everywhere.}$$

OA.4.2 Smooth Noise

We now consider a specification of the noise distribution in the judge's signal which implies that messages that are closer to the judge's observed signal are deemed increasingly more likely to be expert messages, and that no message-signal pairs are ever conclusive evidence of quackery (as in the case of inconsistent messages, in our benchmark setting).

Formally, we change the baseline by assuming that the judge gets a signal $s = \omega + \epsilon$, where ϵ has an even, differentiable density h with full support on $\mathbf{R}$, which we write in the form

$$h(x) = Z^{-1}e^{-V(x)}, \qquad k(x) := V'(x) = -(\ln h)'(x),$$

where V is even, smooth, and strictly convex (so that h is strictly log-concave). Borrowing the terminology of statistics, we refer to k as the *score* of the noise. Strict log-concavity implies that the judge optimally uses a cut-off selection rule: for $m < m'$, the judge selects the speaker sending message m if, and only if, $s < s(m,m')$ for some cut-off $s(m,m')$ in $\mathbf{R} \cup \{\pm\infty\}$. We make the following further Assumptions 1 and 2.[2]

Assumption 1. The noise density is $h = Z^{-1}e^{-V}$, where V is an even function in $C^\infty(\mathbf{R})$ with $V'' > 0$, and the score $k = V'$ is concave on $[0,\infty)$, with $k(x) \to \infty$ as $x \to \infty$.

[2]Assumption 1 has a transparent interpretation in terms of the tails of the noise: they must be intermediate between exponential and Gaussian. Indeed, since $k(0) = 0$, concavity of the score gives $k(x) \leqslant k'(0)x$, and hence $V(x) - V(0) \leqslant k'(0)x^2/2$; unboundedness and monotonicity of the score give $V(x)/x \to \infty$. Consequently,

$$h(x) = o(e^{-c|x|}) \text{ for every } c > 0, \qquad h(x) \geqslant Ce^{-k'(0)x^2/2} \text{ for some } C > 0:$$

the tails of the noise are thinner than every exponential, but no thinner than Gaussian. The Gaussian density, whose score is linear, is the thin-tailed boundary of this class. The class also contains densities with heavier (yet super-exponential) tails: for instance, $V(x) = x\,\mathrm{arsinh}(x) - \sqrt{1+x^2}$, whose score $k(x) = \mathrm{arsinh}(x)$ is concave and unbounded, yields tails of order $e^{-|x|\ln|x| + O(|x|)}$; another convenient family is $V(x) = \gamma x^2/2 + \sum_{j=1}^J \alpha_j \ln\cosh(\beta_j x)$, with $\gamma > 0$ and $\alpha_j, \beta_j \geqslant 0$.

Assumption 2. The state ω is drawn from a distribution G on $[-1,1]$ that is symmetric around 0 and admits a density g in $C^\infty([-1,1])$ with $g>0$ and $g'(m)\leqslant 0$ for m in $(0,1)$.

Theorem 2 in the main text implies the existence of a truth-telling equilibrium in this environment. Under an identification condition on the noise — implied by Assumption 1 — the equilibrium quack strategies are unique, coincide across speakers, and are symmetric around zero.

Proposition OA.4.2. *Suppose that G is atomless and symmetric around 0, and that h is even, positive, and continuous, with the property that, for every $a\neq b$ in $[-1,1]$, the likelihood ratio $s\mapsto h(s-a)/h(s-b)$ has range $(0,\infty)$. Then all saddle points of the truth-telling game use the same pair of quack strategies. The two strategies coincide and are symmetric around 0.*

For a positive quack density f_Q, define its *tilt* by $\tau=\ln(f_Q/g)$.

Proposition OA.4.3. *Suppose that Assumptions 1 and 2 hold. The unique equilibrium quack strategy has a positive, even, Lipschitz continuous density f_Q. Its tilt is Lipschitz continuous and satisfies*

$$-k(m)\leq\tau'(m)\leq 0 \qquad \text{for almost every } m\in(0,1). \tag{9}$$

Under the Bayes cutoff rule, its pure-report payoff is constant on $[-1,1]$. For every $0\leq a<b\leq 1$,

$$-[V(b)-V(a)]<\tau(b)-\tau(a)<0. \tag{10}$$

Consequently, f_Q/g decreases with $|m|$, $f_Q/(gh)$ increases with $|m|$, and f_Q decreases on $(0,1]$, all strictly. For every t in $(0,1)$,

$$\int_{-t}^{t} f_Q(m)\,\mathrm{d}m > G(t)-G(-t).$$

Proposition OA.4.3 describes pandering on the intensive margin. The quack uses every report but overweights moderate messages relative to the prior. His strategy assigns more probability to every central interval, and his density is strictly unimodal. The extent of pandering is bounded by the posterior held by a judge whose signal is central. The two bounds in (10) compare the quack's density with the prior and with this posterior, whose density is proportional to gh.

The economic mechanism is the one anticipated above. For the quack to be willing to use the entire range of messages, he must be compensated for reports that are, on average, farther from the judge's private signal; the compensation takes the form of cut-offs biased in favor of the more extreme report. For this bias to be consistent with Bayes' rule, the judge must regard extreme reports as relatively less likely to come from the quack — relative, that is, to the prior likelihood of the corresponding states. This compensation takes the form described in the following corollary.

Corollary OA.4.4. *Suppose that Assumptions 1 and 2 hold, and let $s(m,m')$ be the equilibrium cut-off for two messages $m<m'$. Then $s(m,m')$ is finite, and it is displaced from the midpoint of the two messages toward the message of smaller absolute value:*

$$s(m,m')<\frac{m+m'}{2} \quad \text{if } |m|<|m'|, \qquad s(m,m')>\frac{m+m'}{2} \quad \text{if } |m|>|m'|,$$

with $s(m,m')=0$ when $m'=-m$. Moreover, for $0\leqslant m<m'$,

$$0<s(m,m')<\frac{m+m'}{2},$$

and the displacement $\delta := (m + m')/2 - s(m, m')$ *is the unique solution of*

$$\int_{d/2-\delta}^{d/2+\delta} k(u)\,\mathrm{d}u = \tau(m) - \tau(m'), \qquad d := m' - m. \tag{11}$$

Under these hypotheses, Corollary OA.4.4 describes the judge's selection rule. The more extreme of two messages is selected on a strictly larger set of signals than the set on which it is the closer of the two: the judge does favor extremity, and the displacement (11) measures by how much — for fixed $d > 0$ and a small displacement, $\delta \simeq [\tau(m) - \tau(m')]/(2k(d/2))$. This approximation need not apply merely because the messages are close. But she never favors it so far as to select it when her signal is exactly central.

OA.4.3 Proofs

OA.4.3.1 Proof of Proposition OA.4.1

Translation continuity in L^1 makes the displayed β positive; the value bounds follow from the capped-game bounds in the proof of Theorem 2 of the paper, taking the limit and specializing to $\hat{h}(s \mid \theta) = h(s - \theta)$. With an uninformed quack, $u_i(T_i) = u_i$ is constant. Thus (8) becomes $u_i F_Q^i(A) \leq G(A)$ for every Borel A, which is exactly the asserted absolute continuity and density cap. With unequal priors, the Bayes comparison in the proof of (8) retains the corresponding prior-odds factor. □

OA.4.3.2 Proof of Proposition OA.4.2

Fix the expert at truth-telling. First, each identity's equilibrium quack payoff is positive. If, for example, speaker 1 earned zero, every pure report would earn zero, so the judge would select speaker 1 with probability zero for every such report, almost surely against the truthful expert. Speaker 2 could then mimic G and be selected with probability one, making the value $1/2$. But the value is strictly below $1/2$: equality would require the two assignment-specific observation laws at a saddle point to coincide; their report marginals would then force both quack laws to equal G, whereas atomlessness and the full-range condition make $h(s - a)$ and $h(s - b)$ differ on a positive-measure set. The density bound of Proposition OA.4.1 therefore applies to both identities. Write their quack laws as $F_Q^i = r_i G$.

For a pair (r_1, r_2), the two assignment likelihoods at reports (a, b) and signal s, after division by $G(\mathrm{d}a)G(\mathrm{d}b)\,\mathrm{d}s$, are $r_2(b)h(s - a)$ and $r_1(a)h(s - b)$. Hence the quack pair at every saddle point maximizes, up to an immaterial positive factor, the concave Bayes-risk functional

$$\mathcal{B}_G(r_1, r_2) = \iiint \min\{r_2(b)h(s - a), r_1(a)h(s - b)\}\,\mathrm{d}s\,G(\mathrm{d}a)G(\mathrm{d}b) \tag{12}$$

over nonnegative densities satisfying $\int r_i\,\mathrm{d}G = 1$.

Let (r_{10}, r_{20}) and (r_{11}, r_{21}) arise at two saddle points. Their midpoint is also a maximizer. Equality in the pointwise concavity of the minimum implies

$$\begin{aligned}[r_{20}(b)h(s - a) - r_{10}(a)h(s - b)] \\ \times [r_{21}(b)h(s - a) - r_{11}(a)h(s - b)] \geqslant 0\end{aligned} \tag{13}$$

for $G \otimes G \otimes$ Leb-almost every (a, b, s).

For each component, the positive supports of the two maximizers coincide modulo G-null sets. Indeed, if $r_{10}(a) > 0 = r_{11}(a)$ on a set of positive measure, choose b with $r_{21}(b) > 0$. The second factor in (13) is positive. If $r_{20}(b) = 0$, the first is negative; if $r_{20}(b) > 0$, the full-range condition makes the first take both signs on positive-measure sets of signals. Either case contradicts (13). Reversing the two maximizers, and then the two identities, proves the claim.

Take a and b in the respective common positive supports. If

$$\frac{r_{10}(a)}{r_{20}(b)} \neq \frac{r_{11}(a)}{r_{21}(b)},$$

continuity and the full-range condition give an open set of signals on which $h(s-a)/h(s-b)$ lies strictly between the two thresholds. The factors in (13) then have opposite signs. The thresholds must therefore agree almost everywhere. Separation of variables gives

$$\frac{r_{10}(a)}{r_{11}(a)} = \frac{r_{20}(b)}{r_{21}(b)} = c$$

almost everywhere on the positive supports, and normalization gives $c = 1$. Thus the pair of equilibrium quack strategies is unique.

Finally, exchanging the speaker labels and reflecting all reports around zero both preserve (12). Applied to the unique pair, exchange gives $r_1 = r_2$ and reflection gives $r_i(a) = r_i(-a)$ almost everywhere. □

OA.4.3.3 Payoffs

For $d > 0$ and ℓ in $\mathbf{R}$, let $z(d, \ell)$ solve

$$\ell - V(z) + V(z + d) = 0, \tag{14}$$

and set $P(d, \ell) = H(z(d, \ell))$, where H is the noise cdf. The score is increasing and unbounded, so the solution is unique. For an even Lipschitz function τ, define its *report payoff* by

$$Q(a) = \int_{-1}^{1} g(b) P(|a - b|, \tau(a) - \tau(b)) \, \mathrm{d}b. \tag{15}$$

The value assigned at $b = a$ is immaterial. The function Q is continuous by dominated convergence. If $f = ge^{\tau}$ is a density, $Q(a)$ is the quack's payoff from report a under the Bayes cutoff rule. Put

$$A(d, \ell) = -P_{\ell}(d, \ell) = \frac{h(z(d, \ell))}{k(z(d, \ell) + d) - k(z(d, \ell))} > 0.$$

Implicit differentiation gives

$$P_d = -Ak(z + d), \qquad z(d, -\ell) = -z(d, \ell) - d, \qquad A(d, -\ell) = e^{\ell} A(d, \ell). \tag{16}$$

Suppose $-k \leq \tau' \leq 0$ almost everywhere on $(0, 1)$. Evenness gives $|\tau(a) - \tau(b)| \leq k(1)|a - b|$. Dividing (14) by $d = |a - b|$ shows that the average of k over $[z, z + d]$ lies in $[-k(1), k(1)]$. Hence z and $z + d$ are in $[-3, 3]$. Since $k' > 0$ on this interval, $A(d, \tau(a) - \tau(b)) \leq C/d$. Write $q = \tau'$ and

$$(\mathcal{L}_g q)(a) = \int_{-1}^{1} A(|a - b|, \tau(a) - \tau(b)) \{q(a) - q(b)\} g(b) \, \mathrm{d}b.$$

If τ is C^2 near an interior point a, this integral is absolutely convergent. Translating $b = a + u$ before differentiating gives

$$Q'(a) = g(1)R(a) - (\mathcal{L}_g q)(a) + N(a), \tag{17}$$

where

$$R(a) = P(1+a, \tau(a)-\tau(1)) - P(1-a, \tau(a)-\tau(1)), \qquad N(a) = \int_{-1}^{1} P(|a-b|, \tau(a)-\tau(b)) g'(b)\, \mathrm{d}b.$$

Alternatively, differentiating at fixed b gives

$$\begin{aligned} Q'(a) = g(a)J(q(a)) &\\ &+ \int_{-1}^{a} g(b)A(a,b)\{-k(z_{ab} + a - b) - q(a)\}\, \mathrm{d}b \\ &+ \int_{a}^{1} g(b)A(a,b)\{k(z_{ab} + b - a) - q(a)\}\, \mathrm{d}b, \end{aligned} \tag{18}$$

where $A(a,b) = A(|a-b|, \tau(a)-\tau(b))$, $z_{ab} = z(|a-b|, \tau(a)-\tau(b))$, and $J(u) = 2H(k^{-1}(-u)) - 1$. The limits of P as $b \uparrow a$ and $b \downarrow a$ are, respectively, $H(k^{-1}(-q(a)))$ and $H(k^{-1}(q(a)))$. Their difference gives the jump term. Local C^2 regularity makes each expression in braces $O(|a-b|)$, uniformly for a in a smaller neighborhood. Thus the integrands in (18) are bounded near the diagonal. Away from it, Lipschitz continuity suffices for differentiation under the integral. For (17), the translated difference quotient of $\tau(a) - \tau(a+u)$ is bounded by $C|u|$ near $u = 0$. This cancels the bound $C/|u|$ on A. Away from $u = 0$, the difference quotients are bounded and converge for almost every u, since τ is Lipschitz. Dominated convergence and the endpoint terms give (17). No additional smoothness outside the neighborhood of a is required.

Lemma OA.4.5. *Suppose τ is even and Lipschitz continuous, with $-k \le \tau' \le 0$ almost everywhere on $(0,1)$. On any open interval I contained in $(0,1)$ where τ is constant, $Q' < 0$. On any such interval where $\tau + V$ is constant, $Q' > 0$.*

Proof. For $0 < a < x$, put $d = x - a$, $e = x + a$, and $\ell = \tau(a) - \tau(x)$. Integrating the derivative bounds $-k \le \tau' \le 0$ from a to x gives $0 \le \ell \le V(x) - V(a)$. Write $z(d, \ell) = -v_d$ and $z(e, \ell) = -v_e$, and put $t_d = d - v_d$, $t_e = e - v_e$. The cutoff equation gives

$$\frac{d}{2} \le v_d \le v_e \le x, \qquad -a \le t_d < t_e, \qquad t_e \ge a, \qquad t_e - t_d \le 2a.$$

For example, both cutoffs are at most x because $V(x) - V(d-x) = V(x) - V(e-x) = V(x) - V(a) \ge \ell$. At $v = v_d$, we have $|e - v| \ge |d - v|$, so the equation for v_e has a weakly smaller left side. Hence $v_e \ge v_d$. At $v = v_d + 2a$, its left side exceeds that for v_d, so $v_e < v_d + 2a$. These comparisons prove the displayed bounds. Since $h(v_e) \le h(v_d)$ and $k(v_e) + k(t_e) > k(v_d) + k(t_d) > 0$, they imply

$$A(|a-x|, \tau(a)-\tau(x)) > A(a+x, \tau(a)-\tau(x)), \tag{19}$$

$$P(|a-x|, \tau(a)-\tau(x)) \ge P(a+x, \tau(a)-\tau(x)). \tag{20}$$

For $x < a$, exchange a and x and use (16); the probability comparison follows from $P(d, -\ell) = H(-t_d)$. Thus these orders hold for all positive $a \ne x$. They give $R(a) \le 0$ and $N(a) \le 0$.

If τ is constant on I, then $q(a) = 0$ for a in I. Pairing the contributions at $b = x$ and $b = -x$, using that g and τ are even and q is odd, gives

$$(\mathcal{L}_g q)(a) = -\int_0^1 \{A(|a-x|, \tau(a) - \tau(x)) - A(a+x, \tau(a) - \tau(x))\} q(x) g(x) \, \mathrm{d}x \geq 0.$$

Equality forces $q = 0$ almost everywhere. In that case τ is constant everywhere and $R(a) < 0$. Equation (17) therefore gives $Q'(a) < 0$ on I.

If $\tau + V$ is constant on I, then $q(a) = -k(a)$ there. Pair $b = x$ and $b = -x$ in (18). For $x < a$, apply the preceding cutoff bounds with a and x exchanged. The identity $z(d, -\ell) + d = -z(d, \ell)$ then makes the two braces $k(a) - k(v_d)$ and $k(a) - k(v_e)$, with $v_d, v_e \leq a$. Both are nonnegative. For $x > a$, the paired expression is

$$(A_d - A_e)\{k(t_d) + k(a)\} + A_e\{2k(a) + k(t_d) - k(t_e)\} \geq 0,$$

where $A_d = A(d, \ell)$ and $A_e = A(e, \ell)$. Indeed, $t_d \geq -a$. Since k is odd and concave on $[0, \infty)$, k' is even and nonincreasing on $[0, \infty)$. Its integral over an interval of length at most $2a$ is no larger than its integral over $[-a, a]$. Thus $k(t_e) - k(t_d) \leq 2k(a)$. The jump term is $g(a)\{2H(a) - 1\} > 0$, so $Q'(a) > 0$ on I. □

OA.4.3.4 Proof of Proposition OA.4.3

Consider the positive, even functions r with $\int r \, \mathrm{d}G = 1$ and

$$e^{-[V(b) - V(a)]} r(a) \leq r(b) \leq r(a), \qquad 0 \leq a < b \leq 1. \tag{21}$$

This set is nonempty, since it contains $r = 1$. Writing $D = V(1) - V(0)$, normalization gives $1 \leq r(0) \leq e^D$ and $e^{-D} \leq r(a) \leq e^D$. For $0 \leq a < b \leq 1$, they also give $0 \leq r(a) - r(b) \leq e^D k(1)(b - a)$. Evenness extends this Lipschitz bound to $[-1, 1]$. The bounds, normalization, and (21) are preserved under uniform limits. The set is therefore compact in the uniform norm. The functional $r \mapsto \mathcal{B}_G(r, r)$ in (12) is continuous, so it has a maximizer on this set. Fix one and write $f = gr$ and $\tau = \ln r$. Condition (21) is equivalent to $-k \leq \tau' \leq 0$ almost everywhere on $(0, 1)$.

Define $v = -\tau'$ on $(0, 1)$, so that $0 \leq v \leq k$ and

$$r(a) = \frac{\exp\{-\int_0^{|a|} v(x) \, \mathrm{d}x\}}{\int_{-1}^1 g(b) \exp\{-\int_0^{|b|} v(x) \, \mathrm{d}x\} \, \mathrm{d}b}.$$

Let Q be the corresponding report payoff and put

$$\mu = \int_{-1}^1 f(a) Q(a) \, \mathrm{d}a, \qquad S(x) = \int_x^1 f(a)\{Q(a) - \mu\} \, \mathrm{d}a.$$

The function S is continuously differentiable. Evenness gives $S(0) = S(1) = 0$. For any measurable w with $0 \leq w \leq k$, replacing v by $v + t(w - v)$, $0 \leq t \leq 1$, defines feasible densities r_t. The derivative of their logarithms at $t = 0$ is

$$-\int_0^{|a|} (w - v)(x) \, \mathrm{d}x + \int_{-1}^1 f(b) \int_0^{|b|} (w - v)(x) \, \mathrm{d}x \, \mathrm{d}b.$$

For distinct reports, the two likelihoods in (12) are equal at only one signal. The report diagonal has zero $G \otimes G$ measure. Thus ties have zero measure under $G \otimes G \otimes \text{Leb}$. Differentiation of the minimum, followed by dominated convergence, gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{B}_G(r_t, r_t)\Big|_{t=0+} = 2\int_{-1}^{1} Q(a)\,\frac{\mathrm{d}r_t(a)}{\mathrm{d}t}\Big|_{t=0} g(a)\,\mathrm{d}a = -4\int_0^1 (w-v)(x)S(x)\,\mathrm{d}x \leq 0.$$

Domination follows from the common bounds on r_t and their derivatives and from integrability of h. Taking $w = 0$ on $\{S > 0\}$ and $w = v$ elsewhere shows that $v = 0$ almost everywhere on $\{S > 0\}$. Taking $w = k$ on $\{S < 0\}$ and $w = v$ elsewhere gives $v = k$ almost everywhere there. Thus τ is constant on each component of $\{S > 0\}$, and $\tau + V$ is constant on each component of $\{S < 0\}$.

If S has a positive maximum at c, then c is in $(0, 1)$ and Lemma OA.4.5 gives $Q'(c) < 0$. Since $S'(c) = -f(c)\{Q(c) - \mu\} = 0$, we have $S''(c) = -f(c)Q'(c) > 0$, a contradiction. Here f and Q are differentiable near c because τ is constant there. A negative minimum similarly gives $S''(c) = -f(c)Q'(c) < 0$, again a contradiction. Consequently $S = 0$, and hence $Q = \mu$ on $[0, 1]$ and, by symmetry, on $[-1, 1]$.

Use the Bayes cutoff rule off the report diagonal and select either speaker with probability $1/2$ on the diagonal. This rule is measurable and optimal against (r, r). Since G is atomless, the diagonal has zero probability against every pure quack report. Each report earns μ, so every quack deviation, including an atom, earns μ. Thus these strategies form a saddle point of the truth-telling game. Proposition OA.4.2 identifies f with the unique equilibrium quack density. This proves positivity, Lipschitz continuity, equalization, and (9).

The inequalities in (21) are strict for $a < b$. Otherwise, either τ or $\tau + V$ would be constant on (a, b), contrary to Lemma OA.4.5 and equalization. This proves (10). It makes f/g strictly decreasing and $f/(gh)$ strictly increasing in $|a|$. Since g is nonincreasing on $[0, 1]$, f is strictly decreasing there. The G-average of f/g is one, so its average on each proper central interval is larger than one. This gives the central-mass comparison. □

OA.4.3.5 Proof of Corollary OA.4.4

The judge selects the speaker sending m' if, and only if, $h(s - m')/h(s - m) > e^{\tau(m')-\tau(m)}$. The left-hand side is strictly increasing in s, because $V'' > 0$ makes the score increasing, and its range is $(0, \infty)$, because the score is unbounded; so the cut-off exists and is finite. Evenness of h makes the left-hand side equal to 1 at the midpoint, and Proposition OA.4.3 makes the right-hand side smaller than 1 when $|m'| > |m|$: the speaker sending m' is then selected at the midpoint, so the cut-off lies strictly below it. The two remaining cases follow by symmetry, the tilt τ being even. For $0 \leqslant m < m'$, the lower bound in (10) makes the right-hand side larger than $h(m')/h(m)$, which is the value of the left-hand side at $s = 0$; the speaker sending m is therefore selected at the central signal, so the cut-off lies strictly above 0. Finally, writing $s = (m + m')/2 - \delta$ gives $s - m = d/2 - \delta$ and $s - m' = -(d/2 + \delta)$, so that indifference reads $V(d/2 - \delta) - V(d/2 + \delta) = \tau(m') - \tau(m)$, which is (11); the left-hand side of (11) vanishes at $\delta = 0$ and has derivative $k(d/2 + \delta) + k(d/2 - \delta) > 0$, so the solution is unique. □

OA.5 Polite Talk

Consider the benchmark formulation, in which the state is distributed uniformly, the judge's signal has uniform noise, and the judge has a symmetric prior about the speakers' expertise types.

For convenience, in this section we assume $\bar{\epsilon} < 1/4$, so that the judge's private signal is sufficiently precise. Now consider the game in which the judge randomly picks one of the two speakers to be the first to talk, publicly, after which the other speaker gets a chance to do the same. The judge waits until both speakers have talked before selecting the winner.

There exists an equilibrium of this "polite talk" game in which the expert reports the true state, regardless of when he is called on to speak.

The quack's strategy now involves both a distribution F_Q^1 in $\Delta\mathcal{M}$, in case he is to speak first, and a transition kernel F_Q^2 on $\mathcal{M} \times \mathcal{M}$, in case he is called upon to speak second, defining a measure $F_Q^2(\cdot \mid \omega)$ on $\mathcal{M}$ specifying what message he sends given the message $m_1 = \omega$ sent by the expert. Note that, as long as the expert's strategy is to tell the truth, every m_1 in $[-1, 1]$ is on path, and no refinement is needed.

When speaking second, the quack first learns the state (from the report of the first speaker, whom he knows to be the expert). If the quack repeats the initial report, he knows his message will certainly be consistent. The judge, however, would suspect their expertise, regardless of consistency. A consistent message by the first speaker, instead, would carry more persuasion. In fact, Proposition OA.5.1 describes an equilibrium in which, conditional on the first speaker being consistent, the second speaker is never selected.[3]

Proposition OA.5.1. *There exists an equilibrium with the following strategies. The expert reports the state truthfully, whether he speaks first or second.*

If the quack speaks first, his messaging strategy $F_Q^1(\cdot) = \mathcal{U}[-(1-2\bar{\epsilon}), 1-2\bar{\epsilon}]$. *If the quack speaks second, he uses as messaging strategy* $F_Q^2(\cdot \mid m_1) = \mathcal{U}[\max\{-1, m_1 - 2\bar{\epsilon}\}, \min\{1, m_1 + 2\bar{\epsilon}\}]$.

The judge selects the first speaker if, and only if, his message is consistent.

Plainly, when speaking first, the quack wants to maximize the probability of consistency, which F_Q^1 achieves. When speaking second, his choice is irrelevant, as he anticipates that he won't be selected. A first-moving expert is selected with certainty by reporting truthfully. A second-moving expert cannot affect selection, so truthful reporting is a weak best response. It remains to check the judge's selection rule. That is, when facing two consistent messages with m_1 in $[-(1-2\bar{\epsilon}), 1-2\bar{\epsilon}]$, the judge must prefer opting for the first speaker. The judge compares

$$\frac{1}{2}\frac{1}{2\bar{\epsilon}}\frac{1}{2(1-2\bar{\epsilon})} \gtrless \frac{1}{2}\frac{1}{2\bar{\epsilon}}\frac{1}{4\bar{\epsilon}}.$$

In both cases, one of the two speakers told the truth (density 1/2); and, conditional on the truth, the consistent signal's density is $\frac{1}{2\bar{\epsilon}}$. The density of the quack who speaks first is $\frac{1}{2(1-2\bar{\epsilon})}$, whereas $\frac{1}{4\bar{\epsilon}}$ if he speaks second; and so (because $\bar{\epsilon} < 1/4$), the second event is more likely, and the judge selects the first of the two speakers.

While there are other equilibria of this sequential talk game, every equilibrium in which the expert tells the truth yields the same payoff. (Because the game between quack and judge is zero-sum.) In our equilibrium, the quack is selected if, and only if, he is chosen to be first (which occurs with probability 1/2), and his message –say, message 0– happens to be consistent. His payoff is $\frac{1}{2}\int_{-\bar{\epsilon}}^{\bar{\epsilon}} \mathbf{P}(s)\,\mathrm{d}s = \frac{\bar{\epsilon}}{2}$. In contrast, in the simultaneous communication game, the quack's payoff is $\bar{\epsilon}/2 - \bar{\epsilon}^2/6$, confirming that the quack is better off (and the judge worse off) with

[3]Elements of the equilibrium specification in Proposition OA.5.1 are largely arbitrary: given that a quack that comes second is never selected, he might as well randomize over all possibly consistent messages. If the quack comes first, he needs to make sure to maximize the probability of consistency, that is, his strategy must be concentrated on $[-(1-2\bar{\epsilon}), 1-2\bar{\epsilon}]$, but here as well, there is some leeway in the density of F_Q^1.

sequential communication. This is not surprising, given the revelation principle, and the zero-sum nature of the interaction between the judge and the quack.

The key insight is that sequential communication destroys the quack's *ex ante* symmetry with the expert. Once the second speaker can condition on the first, the judge must discipline the second speaker harshly, which mechanically raises the quack's expected payoff and lowers screening power. In this sense, simultaneity is useful in that it equalizes the strategic positions of the senders and makes relative credibility informative.

OA.6 One Speaker

Suppose there is a single speaker, who is a quack with probability q in $(0,1)$. The judge, upon receiving a message m in $\mathcal{M}$, chooses between selecting the speaker, which yields a payoff of 1 if he is an expert, and 0 otherwise, or taking an outside option, with fixed value $\mathcal{U}$ in $(0,1)$. Assume $0 < \bar{\epsilon} < 1/2$. The model is otherwise unchanged from our benchmark: the state ω is uniformly distributed on $[-1,1]$, and the judge observes a signal $s = \omega + \epsilon$, with ϵ uniform on $[-\bar{\epsilon}, \bar{\epsilon}]$. We focus on the (perfect Bayesian) equilibrium that is preferred by the judge.

The characterization of the judge's best equilibrium depends on the value of the outside option. We let $\mathcal{V} := \frac{1-q}{q}\frac{1-\mathcal{U}}{\mathcal{U}}$ denote the expected gain from picking an expert over the outside option, relative to the expected loss from picking a quack over the outside option. A larger $\mathcal{V}$ means a stronger preference for selecting the speaker over the outside option, all else equal.

Proposition OA.6.1. *There is an equilibrium that is best for the judge in which the expert type reports the state truthfully. As for the quack type, we distinguish two settings, depending on the outside option value:*

If $\mathcal{V} \geqslant \bar{\epsilon}$, the quack uses a messaging strategy F_Q given by $\mathcal{U}[-\bar{m}, \bar{m}]$. If $\mathcal{V}$ is in $[\bar{\epsilon}, \frac{\bar{\epsilon}}{1-2\bar{\epsilon}})$, $\bar{m} = \bar{\epsilon}/\mathcal{V}$. If $\mathcal{V} \geq \frac{\bar{\epsilon}}{1-2\bar{\epsilon}}$, any $\bar{m}$ in $[\bar{\epsilon}/\mathcal{V}, 1-2\bar{\epsilon}]$ can be used. This interval is a singleton at equality.

If $\mathcal{V} < \bar{\epsilon}$, the quack's messaging strategy F_Q is supported on $[-1,1]$, with density

$$f_Q(m) = \begin{cases} \frac{\mathcal{V}}{2\bar{\epsilon}}, \text{ for } |m| \ \leq 1 - 2\bar{\epsilon}z, \\ \frac{\mathcal{V}}{\sqrt{(1-|m|)^2 + 4\bar{\epsilon}^2(1-z^2)}}, \text{ for } |m| \ \geq 1 - 2\bar{\epsilon}z, \end{cases}$$

where z is the unique solution in $(0,1)$ to

$$\frac{1}{\bar{\epsilon}} - \frac{1}{\mathcal{V}} = 2z - 2\tanh^{-1}(z). \tag{22}$$

Figure 1 illustrates the quack-type speaker's equilibrium messaging strategy, in the lemon-dropping setting (when $\mathcal{U}$ is small, 2/3) and in the cherry-picking setting (when $\mathcal{U}$ is large, 4/5). In the former case, the quack "panders" to moderate messages, by using a truncated support of messages. In the latter case, the quack "panders" to extreme messages, by over-weighting them relative to moderate ones.

To understand the equilibrium, consider the judge's problem. Assuming the expert reveals the state truthfully, and that the quack uses a messaging strategy with density f_Q, when the judge sees a private signal s and a message m that is consistent with s, she compares

$$\mathcal{U} \gtrless \frac{\frac{1-q}{2}\frac{1}{2\bar{\epsilon}}}{\frac{1-q}{2}\frac{1}{2\bar{\epsilon}} + q f_Q(m)\mathbf{P}(s)}, \tag{23}$$

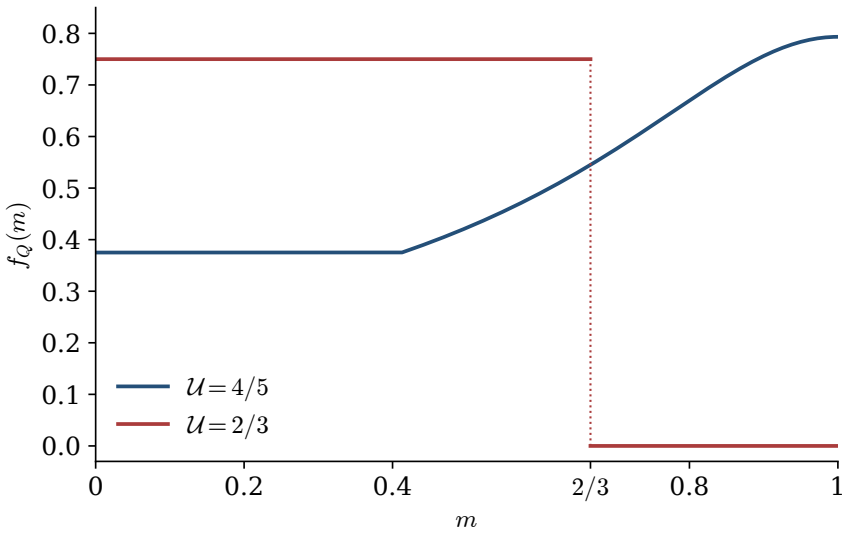


Figure 1: Quack densities for $q = 1/2$, $\bar{\epsilon} = 1/3$, and two outside options.

where the right-hand side is the posterior probability that the speaker is an expert, given (m, s).[4] Conditional on the speaker being the expert type, every pair of consistent message and signal arises with the same probability. In contrast, if the speaker is the quack type, message m and signal s are drawn independently, and their joint probability is the product of their marginals. In particular, fixing the value of $f_Q(m)$ and maintaining consistency, more extreme combinations of m and s are less likely to arise, since $\mathbf{P}(s)$ is nonincreasing in $|s|$. A consequence is that extreme messages m, when consistent, tend to be strong signals that the speaker is an expert.

When the outside option is low, so that $\mathcal{V} \geqslant \bar{\epsilon}/(1-2\bar{\epsilon})$, the judge is willing to select the speaker whenever his report is consistent. The quack can then use any of the stated supports, all of which maximize the probability of consistency. There are multiple support choices when the inequality is strict. In the intermediate case, $\mathcal{V} \in [\bar{\epsilon}, \bar{\epsilon}/(1-2\bar{\epsilon}))$, the judge is indifferent on the central range of signals. Her tie-breaking compensates the quack for using reports with lower probabilities of consistency. Thus consistency alone determines the quack's incentives in the low outside-option case, but not in the intermediate case.

In contrast, in the cherry-picking setting of high outside options ($\mathcal{V} < \bar{\epsilon}$), the judge requires a very strong signal of expertise in order to select the speaker. For a nonnegative report m, $\mathbf{P}(s)$ is nonincreasing over its consistency interval. The judge's payoff from selecting the speaker therefore weakly increases with s on that interval. On the right tail, selection takes the form of an upper cutoff $s(m)$. Negative reports use the reflected rule. For $|s| \leq 1 - \bar{\epsilon}$, $\mathbf{P}(s)$ is constant: hence, if the judge is indifferent for such a signal between the outside option and the speaker, she can adjust the probability of selecting the speaker for signals over this range such that, in expectation, the speaker is indifferent over different messages. Instead for $s(m) > 1 - \bar{\epsilon}$, the cut-off $s(m)$ is required to increase with m, but not as fast as m, since higher messages are less likely to be consistent. Hence, because the judge is indifferent at the cut-off between the outside option and the speaker, higher messages in the support must be sent with higher probability $f_Q(m)$, to offset

[4]The numerator is the probability that the speaker is an expert $(1-q)$, times the density of m given that the expert tells the truth $(1/2)$, times the density of the signal given m $(\frac{1}{2\bar{\epsilon}})$. The denominator additionally includes the probability that the speaker is a quack, times the probability that the quack sends message m, times the probability of s.

the decrease in $\mathbf{P}(s(m))$. So, the quack "panders to the extreme."

In addition to verifying that these strategies form an equilibrium, the proof of Proposition OA.6.1 shows that this equilibrium attains the highest payoff for the judge. As before, we establish this by arguing that the payoff attained in our described equilibrium yields the same payoff as what the judge obtains under commitment. This equivalence is not immediate from our previous analysis, since in the one-speaker case with an outside option, the game between the quack and the judge is no longer zero-sum, fixing the expert's strategy at truth-telling.

OA.6.1 Proof of Proposition OA.6.1

By the revelation principle, it suffices to bound the judge's payoff in a direct mechanism. Relax the expert's incentive constraints and fix his report at the true state. Let α be the quack's selection probability when he discloses his type, and let $\phi(m, s)$ be the selection probability after he claims expertise and reports m. Set $\phi(m, s) = 0$ when $|m - s| > \bar{\epsilon}$. The quack's incentive constraints are

$$\int_{m-\bar{\epsilon}}^{m+\bar{\epsilon}} \mathbf{P}(s)\phi(m, s)\,\mathrm{d}s \leq \alpha \qquad (-1 \leq m \leq 1). \tag{24}$$

Let H be the signal cdf, and write

$$K(m) = H(m + \bar{\epsilon}) - H(m - \bar{\epsilon})$$

for the *consistency probability* of report m. Since $K \leq \bar{\epsilon}$, there is no gain from choosing $\alpha > \bar{\epsilon}$.

For fixed $\alpha \in [0, \bar{\epsilon}]$, let $A(\alpha)$ denote the *maximal average selection probability of the truthful expert* subject to (24). Mixing two feasible selection rules proves that A is concave. Net of the outside option and divided by $q\mathcal{U}$, the judge's objective is

$$\mathcal{V}A(\alpha) - \alpha.$$

For $m \geq 0$, the density $\mathbf{P}(s)$ is nonincreasing on $[m - \bar{\epsilon}, m + \bar{\epsilon}]$, since $\bar{\epsilon} < 1/2$. An upper interval therefore maximizes its length subject to (24). The optimal cutoff is

$$s_\alpha(m) = \begin{cases} m - \bar{\epsilon}, & K(m) \leq \alpha, \\ H^{-1}(H(m + \bar{\epsilon}) - \alpha), & K(m) > \alpha. \end{cases}$$

Negative reports use the reflected rule. Hence

$$A(\alpha) = \int_0^1 \frac{m + \bar{\epsilon} - s_\alpha(m)}{2\bar{\epsilon}}\,\mathrm{d}m.$$

For $0 < \alpha < \bar{\epsilon}$, differentiation gives

$$A'(\alpha) = \frac{1}{2\bar{\epsilon}} \int_{\{m \in [0,1]: K(m) > \alpha\}} \frac{\mathrm{d}m}{\mathbf{P}(s_\alpha(m))}. \tag{25}$$

Suppose $\bar{\epsilon}/2 \leq \alpha < \bar{\epsilon}$. The binding reports form $[0, \bar{m})$, where $K(\bar{m}) = \alpha$ and $\bar{m} \in (1 - 2\bar{\epsilon}, 1]$. Their cutoffs lie in the constant-density region, so $A'(\alpha) = \bar{m}/\bar{\epsilon}$. Thus, if

$$\bar{\epsilon} \leq \mathcal{V} < \frac{\bar{\epsilon}}{1 - 2\bar{\epsilon}},$$

the optimum satisfies $\bar{m} = \bar{\epsilon}/\mathcal{V}$ and $\alpha = K(\bar{m})$. If $\mathcal{V} \geq \bar{\epsilon}/(1-2\bar{\epsilon})$, the optimum is $\alpha = \bar{\epsilon}$, since $A'(\bar{\epsilon}-) = (1-2\bar{\epsilon})/\bar{\epsilon}$.

If $\mathcal{V} < \bar{\epsilon}$, the optimum has $0 < \alpha < \bar{\epsilon}/2$, and every report constraint binds. Put $z = \sqrt{1-2\alpha/\bar{\epsilon}}$. The cutoff lies in the constant-density region for $0 \leq m \leq 1-2\bar{\epsilon}z$ and in the right tail for larger m. Using the signal density

$$\mathbf{P}(s) = \begin{cases} 1/2, & |s| \leq 1-\bar{\epsilon}, \\ (1+\bar{\epsilon}-|s|)/(4\bar{\epsilon}), & 1-\bar{\epsilon} < |s| \leq 1+\bar{\epsilon}, \\ 0, & |s| > 1+\bar{\epsilon}, \end{cases}$$

equation (25) becomes

$$A'(\alpha) = \frac{1}{\bar{\epsilon}} - 2z + 2\tanh^{-1}(z).$$

The first-order condition $\mathcal{V}A'(\alpha) = 1$ is (22). Its right-hand side is decreasing from zero to $-\infty$ as z increases from zero to one, so the solution exists and is unique. Concavity proves optimality. The optimal quack payoff is therefore $\alpha = (1-z^2)\bar{\epsilon}/2$.

We next implement these bounds without commitment. Reject inconsistent reports and use the cutoff s_α for nonnegative reports, with reflection for negative reports. When $\mathcal{V} \in [\bar{\epsilon}, \bar{\epsilon}/(1-2\bar{\epsilon}))$, set $\bar{m} = \bar{\epsilon}/\mathcal{V}$ and let the quack use $\mathcal{U}[-\bar{m}, \bar{m}]$. On this support, $f_Q = \mathcal{V}/(2\bar{\epsilon})$. By (23), the judge is indifferent on the constant-density region and selects the speaker on the tails whenever consistent. The cutoff s_α is therefore optimal. Reports outside the quack's support identify the expert and are selected whenever consistent. Each report in the support earns $\alpha = K(\bar{m})$; all others earn at most α.

When $\mathcal{V} \geq \bar{\epsilon}/(1-2\bar{\epsilon})$, take any $\bar{m} \in [\bar{\epsilon}/\mathcal{V}, 1-2\bar{\epsilon}]$ and let the quack use $\mathcal{U}[-\bar{m}, \bar{m}]$. On the support,

$$f_Q(m)\mathbf{P}(s) \leq \frac{1}{4\bar{m}} \leq \frac{\mathcal{V}}{4\bar{\epsilon}}.$$

The judge therefore selects every consistent report. All reports in the quack's support have consistency probability $\bar{\epsilon}$, which is maximal.

When $\mathcal{V} < \bar{\epsilon}$, the Bayes indifference condition at $s_\alpha(m)$ gives

$$f_Q(m) = \frac{\mathcal{V}}{4\bar{\epsilon}\,\mathbf{P}(s_\alpha(m))} \qquad (0 \leq m \leq 1). \tag{26}$$

Extend this density evenly. On the constant-density region it equals $\mathcal{V}/(2\bar{\epsilon})$. On the right tail,

$$\mathbf{P}(s_\alpha(m)) = \frac{\sqrt{(1-m)^2 + 8\bar{\epsilon}\alpha}}{4\bar{\epsilon}},$$

which gives the density in the proposition. Its integral is $\mathcal{V}A'(\alpha) = 1$. The cutoff is Bayes optimal and every quack report earns α.

It remains to verify the expert's incentives. The cutoff s_α is absolutely continuous. Where the report constraint binds, its derivative satisfies, almost everywhere,

$$s'_\alpha(m) = \frac{\mathbf{P}(m+\bar{\epsilon})}{\mathbf{P}(s_\alpha(m))} \in [0,1].$$

Where it does not bind, $s_\alpha(m) = m - \bar{\epsilon}$. Thus both $s_\alpha(m)$ and $m + \bar{\epsilon} - s_\alpha(m)$ are nondecreasing on $[0,1]$. An expert with state $\omega \geq 0$ who reports $m \geq 0$ is selected with probability

$$\frac{\big(\min\{\omega+\bar{\epsilon}, m+\bar{\epsilon}\} - \max\{\omega-\bar{\epsilon}, s_\alpha(m)\}\big)_+}{2\bar{\epsilon}}.$$

If $m \geq \omega$, monotonicity of the cutoff bounds this by the truthful payoff. If $m \leq \omega$, monotonicity of the accepted interval's length gives the same bound.

For $m < 0$, the intersection of the true and reported consistency intervals lies in $[-\bar{\epsilon}, \bar{\epsilon}]$, where $\mathbf{P}(s) = 1/2$. The quack's incentive constraint therefore bounds the expert's deviation payoff by $\alpha/\bar{\epsilon}$. If the truthful report constraint binds, its accepted interval has length at least 2α, since $\mathbf{P}(s) \leq 1/2$. Truth then yields at least $\alpha/\bar{\epsilon}$. If it does not bind, truth yields one. Reflection covers negative states. Truthful reporting is therefore optimal. The resulting equilibrium attains the commitment bound and is best for the judge. □

OA.7 Blackwell Comparison

We present numerical evidence that the experiment generated by a partition equilibrium need not be a garbling of the truth-telling experiment.

Specification. The state is uniform on $\Omega = \{-1, -2/3, -1/3, 0, 1/3, 2/3, 1\}$. Let $\sigma = 0.7$ and let s_j, $j = 0, \ldots, 20$, be the equally spaced points of $[-1 - 3\sigma, 1 + 3\sigma]$. The judge observes a signal on this common grid with conditional law

$$\mathbf{P}(s = s_j \mid \omega) = \frac{\exp(-(s_j - \omega)^2/(2\sigma^2))}{\sum_{\ell=0}^{20} \exp(-(s_\ell - \omega)^2/(2\sigma^2))}.$$

The identity priors are equal. Each speaker can use seven state labels and three distinct cell labels. These ten messages are available in both equilibria. The calculations below use finite-dimensional linear programs and report numerical solutions.

Computed Equilibria. In the computed *truth-telling equilibrium*, the expert reports the state; the quack's equilibrium messaging strategy (the saddle point of the finite zero-sum truth-telling game, computed by linear programming) is

$$r^* = (0.1188,\ 0.1373,\ 0.1563,\ 0.1750,\ 0.1563,\ 0.1373,\ 0.1188),$$

and the judge uses a minimax selection rule, which is Bayes optimal. In the computed *partition equilibrium*, the expert reports only the cell of the partition

$$\{-1, -\tfrac{2}{3}\}, \qquad \{-\tfrac{1}{3}, 0, \tfrac{1}{3}\}, \qquad \{\tfrac{2}{3}, 1\}$$

containing the state; the quack mixes over the three cell-messages with weights

$$w^* = (0.2692,\ 0.4616,\ 0.2692),$$

and the judge uses a minimax selection rule, which is Bayes optimal. To check the equilibrium conditions numerically, choose the judge's tie probabilities jointly with the incentive constraints: w^* is the saddle point of the induced finite zero-sum game, so the quack is indifferent over the three messages and the judge's rule is optimal; a label-symmetric minimax rule makes reporting the state's cell preferable to either alternative at every state, with minimum slack approximately 2.7062×10^{-5} — and deviations to unused messages are deterred by off-path beliefs that assign the deviation to the quack.

The Comparison. Each equilibrium induces an experiment about the state: the conditional law of the ordered report pair and signal. A separating decision problem has 189 actions, indexed by the partition experiment's report pairs and signal values. Its payoffs lie in $[-7, 7]$. Direct evaluation of its Bayes values gives approximately 6.103489 under truthful reporting and 6.124139 under the partition equilibrium. The difference is $0.020650 > 0$. These computed values indicate that the truth-telling experiment does not Blackwell-dominate the partition experiment (Blackwell, 1953).

Details available upon request.

References

Blackwell, D. (1953). "Equivalent Comparisons of Experiments," *Annals of Mathematical Statistics*, **24**, 265–272.

Lehmann, E. L. (1988). "Comparing Location Experiments," *Annals of Statistics*, **16**, 521–533.